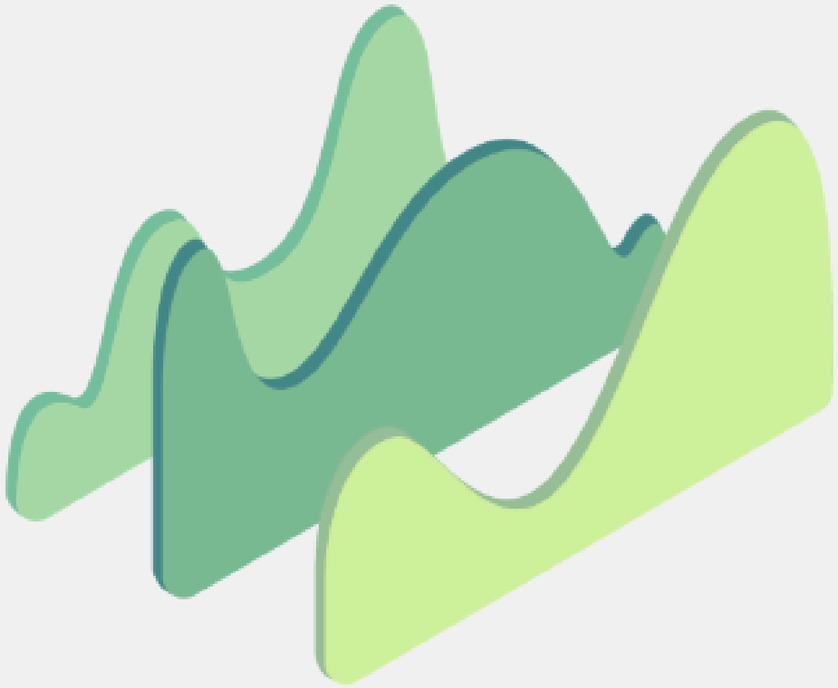
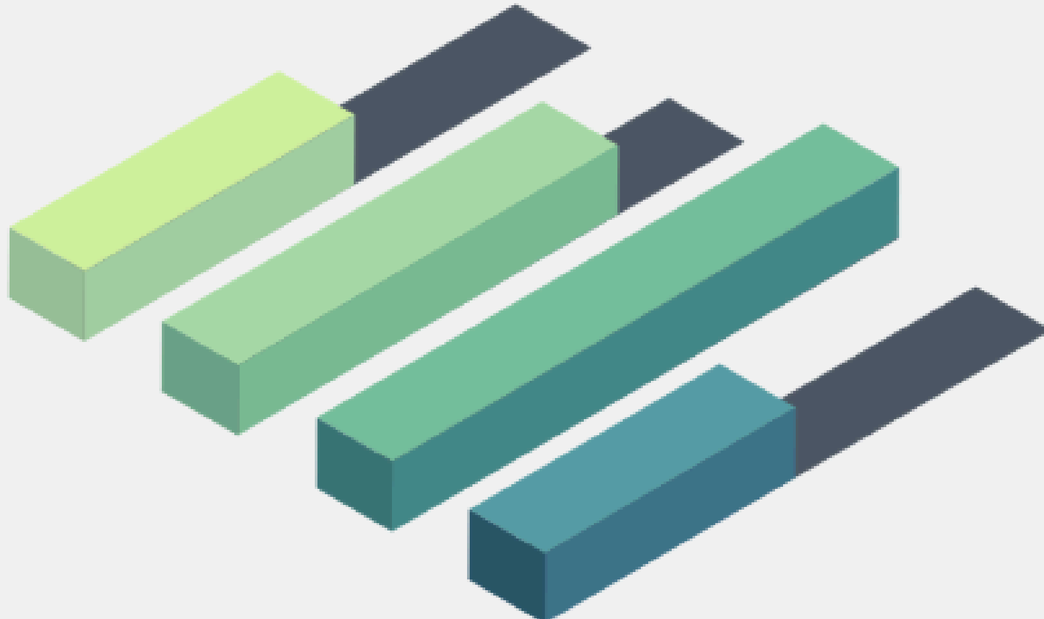
200Lab Education

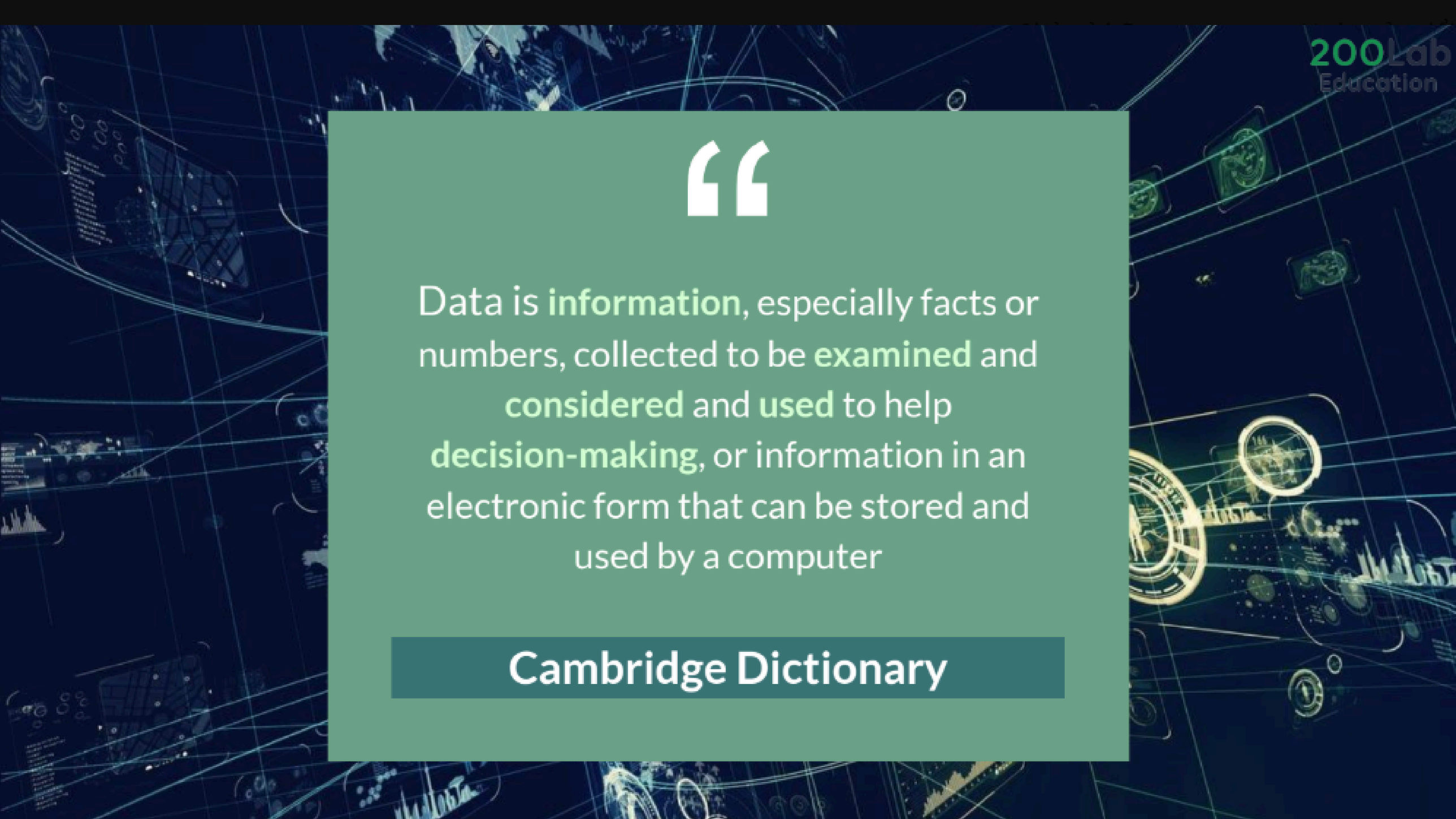
Overview of Database

Bao Tran - Data Mentor





200Lab
Education



Data is **information**, especially facts or numbers, collected to be **examined** and **considered** and **used** to help **decision-making**, or information in an electronic form that can be stored and used by a computer

Cambridge Dictionary

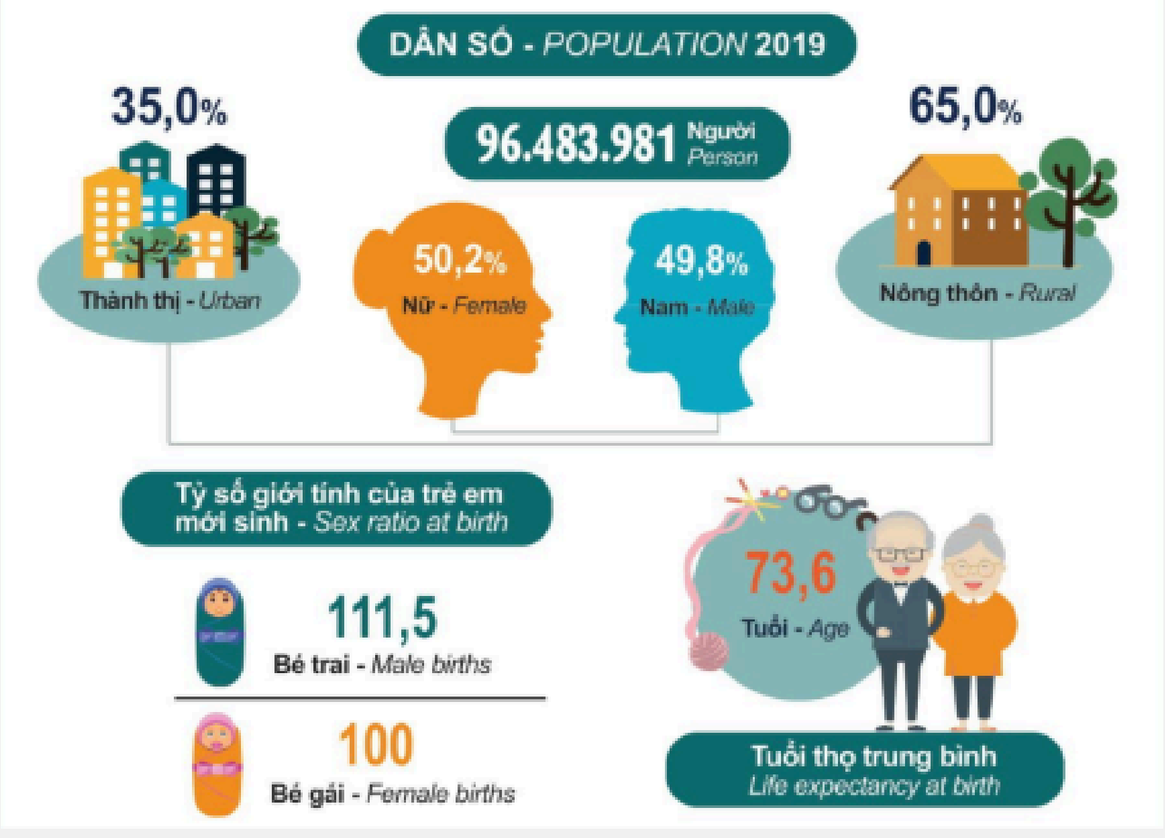
What is Data?

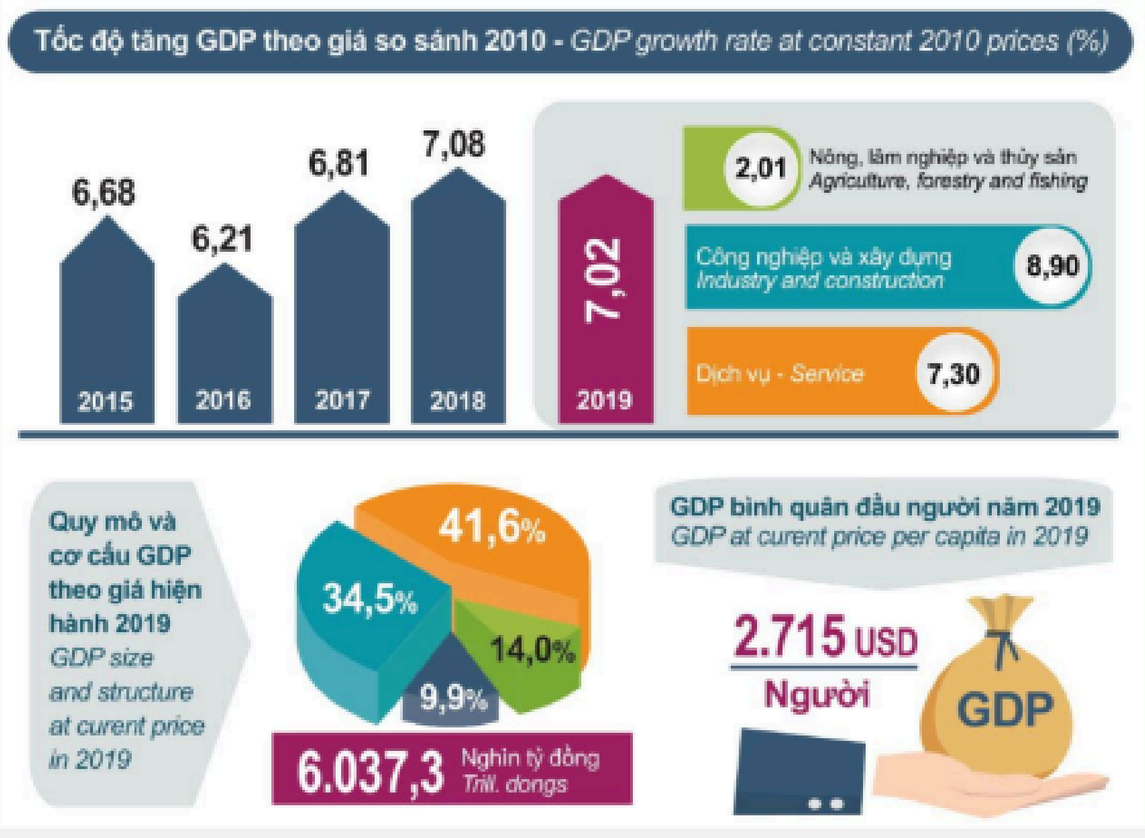
information in an electronic form that can be stored and used by a computer



Examples of Data

Everything can be data and many things can be tracked.







Database Management System

a computerized data-keeping system.

Users of the system are given facilities to perform several kinds of operations on such a system for either manipulation of the data in the database or the management of the database structure itself.

DBMS Classification

	customerName	contactLastName	contactFirstName
1	Atelier graphique	Schmitt	Carine
2	Signal Gift Stores	King	Jean
3	Australian Collectors, Co.	Ferguson	Peter
4	La Rochelle Gifts	Labrune	Janine
5	Baane Mini Imports	Bergulfsen	Jonas
6	Mini Gifts Distributors Ltd.	Nelson	Susan
7	Havel & Zbyszek Co	Piestrzeniewicz	Zbyszek
8	Blauer See Auto, Co.	Keitel	Roland
9	Mini Wheels Co.	Murphy	Julie
10	Land of Toys Inc.	Lee	Kwai

RELATIONAL

A relational database is a logical grouping of data that contains a set of related table spaces and index spaces. In relational database, data is stored in tabular format.



NON-RELATIONAL

Non-relational databases (often called NoSQL databases) are different from traditional relational databases in that they store their data in a non-tabular form.

What is a Table?

A Table is a collection of data where data is logically organized in a **row-and-column format** similar to a spreadsheet. Each row represents a unique record, and each column represents a field in the record.

Column/Field

	customerNumber	customerName	contactLastName	contactFirstName	phone
1	103	Atelier graphique	Schmitt	Carine	40.32.2555
2	112	Signal Gift Stores	King	Jean	7025551838
3	114	Australian Collectors, Co.	Ferguson	Peter	03 9520 4555
4	119	La Rochelle Gifts	Labrune	Janine	40.67.8555
5	121	Baane Mini Imports	Bergulfsen	Jonas	07-98 9555
6	124	Mini Gifts Distributors Ltd.	Nelson	Susan	4155551450
7	125	Havel & Zbyszek Co	Piestrzeniewicz	Zbyszek	(26) 642-7555
8	128	Blauer See Auto, Co.	Keitel	Roland	+49 69 66 90 2555
9	129	Mini Wheels Co.	Murphy	Julie	6505555787
10	131	Land of Toys Inc.	Lee	Kwai	2125557818
11	141	Euro+ Shopping Channel	Freyre	Diego	(91) 555 94 44
12	144	Volvo Model Replicas, Co	Berglund	Christina	0921-12 3555
13	145	Danish Wholesale Imports	Petersen	Jytte	31 12 3555
14	146	Saveley & Henriot, Co.	Saveley	Mary	78.32.5555
15	148	Dragon Souveniers, Ltd.	Natividad	Eric	+65 221 7555
16	151	Muscle Machine Inc	Young	Jeff	2125557413
17	157	Diecast Classics Inc.	Leong	Kelvin	2155551555
18	161	Technics Stores Inc.	Hashimoto	Juri	6505556809
19	166	Handji Gifts& Co	Victorino	Wendy	+65 224 1555

Row/Record

Long vs. Wide Format Table

There are 2 main types of tables:

1. Wide format (human-friendly) contains values that **do not repeat** in the key columns.
2. Long format (machine-readable) contains values that **do repeat** in the key columns.

	contactFirstName	phone	addressLine1	city	state
1	Adrian	+61 2 9495 8555	Monitor Money Building	Chatswood	NSW
2	Akiko	+81 3 3584 0555	2-2-8 Roppongi	Minato-ku	Tokyo
3	Alejandra	(91) 745 6555	Gran Vía, 1	Madrid	[NULL]
4	Alexander	0342-555176	Heerstr. 22	Leipzig	[NULL]
5	Alexander	+7 812 293 0521	2 Pobedy Square	Saint Petersbur	[NULL]
6	Allen	6175558555	7825 Douglas Av.	Brickhaven	MA
7	Ann	(171) 555-0297	35 King George	London	[NULL]
8	Anna	02 9936 8555	201 Miller Street	North Sydney	NSW
9	Annette	61.77.6555	1 rue Alsace-Lorraine	Toulouse	[NULL]
10	Armand	+27 21 550 3555	1250 Pretorius Street	Hatfield	Pretoria
11	Arnold	+63 2 555 3587	15 McCallum Street	Makati City	[NULL]
12	Ben	61-7-3844-6555	31 Duncan St. West End	South Brisbane	Queensland
13	Bradley	+31 20 491 9555	Kingsfordweg 151	Amsterdam	[NULL]
14	Braun	0452-076555	Hauptstr. 29	Bern	[NULL]
15	Brian	2155554369	6047 Douglas Av.	Los Angeles	CA

contactFir	phone	addressl	detailAddress
Adrian	+61 2 9495 8555	address	Monitor Money Building
Adrian	+61 2 9495 8555	city	Chatswood
Adrian	+61 2 9495 8555	state	NSW
Akiko	+81 3 3584 0555	address	2-2-8 Roppongi
Akiko	+81 3 3584 0555	city	Minato-ku
Akiko	+81 3 3584 0555	state	Tokyo
Alejandra	(91) 745 6555	address	Gran Vía, 1
Alejandra	(91) 745 6555	city	Madrid
Alexander	0342-555176	address	Heerstr. 22
Alexander	+7 812 293 0521	address	2 Pobedy Square
Alexander	0342-555176	city	Leipzig
Alexander	+7 812 293 0521	city	Saint Petersburg
Allen	6175558555	address	7825 Douglas Av.
Allen	6175558555	city	Brickhaven
Allen	6175558555	state	MA
Ann	(171) 555-0297	address	35 King George
Ann	(171) 555-0297	city	London
Anna	02 9936 8555	address	201 Miller Street
Anna	02 9936 8555	city	North Sydney
Anna	02 9936 8555	state	NSW
Annette	61.77.6555	address	1 rue Alsace-Lorraine
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Arnold	+63 2 555 3587	address	15 McCallum Street
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Ben	61-7-3844-6555	address	31 Duncan St. West End
Ben	61-7-3844-6555	city	South Brisbane
Ben	61-7-3844-6555	state	Queensland
Bradley	+31 20 491 9555	address	Kingsfordweg 151
Bradley	+31 20 491 9555	city	Amsterdam
Braun	0452-076555	address	Hauptstr. 29
Braun	0452-076555	city	Bern


Thanks!